

SIXTH SEMESTER DIPLOMA EXAMINATION IN ENGINEERING
AND TECHNOLOGY

CONTEMPORARY ELECTRONICS
MODEL QUESTION PAPER – SET-1

Time: 3 hours

Maximum Marks: 75

PART A

I. Answer all the following questions in one word or sentence.

(9 x 1 = 9 Marks)

		Module Outcome	Cognitive level
1	List any two active component in an IC	M1.01	R
2	State the functions of ALU in a microprocessor	M1.02	R
3	Define the term UHD	M2.02	R
4	Name a digital interface for both audio and video	M2.03	R
5	State the function of an inverter	M3.01	R
6	Define the process of converting light (photons) to electricity (voltage)	M3.02	R
7	List two features of VRLA batteries	M3.03	R
8	Expand EVSE	M4.02	R
9	Name any connector type used for electric vehicle charging	M4.03	R

PART B

II. Answer any Eight questions from the following

(8 x 3= 24 Marks)

		Module Outcome	Cognitive level
1	Define Moore's Law	M1.01	R
2	List the features of Microcontrollers makes it to be used in consumer electronics applications	M1.02	R
3	Explain the working principle of Microphone	M2.01	U
4	State the functioning of stereo sound system	M2.01	R
5	Compare various resolution parameters used in video displays	M2.02	U
6	Compare DVD and Blue Ray disc	M2.03	R
7	Explain working of Induction Stoves	M3.01	U

8	Explain precautions while handling, maintenance of lead acid batteries	M3.03	R
9	Draw the block diagram of an electric vehicle charger	M4.02	R
10	Explain different standards of Charging topology	M4.03	U

PART C

III. Answer all questions from the following (6x 7 = 42 Marks)

Module Outcome Cognitive level

1	Draw and Explain the working of Vacuum Tubes	M1.01	U
OR			
2	Compare Microprocessor Vs Microcontrollers	M1.02	R
3	Draw and explain the block diagram of AM Receiver	M2.01	U
OR			
4	Explain various video ports	M2.03	U
5	Define the term tonnage, and explain, how tonnage is calculated for air conditioner	M3.01	U
OR			
6	With block diagram explain on-grid solar energy system	M3.02	U
7	Draw the block diagram and compare online and offline UPS	M3.02	R
OR			
8	Define efficiency, Ampere hour efficiency and watt hour efficiency of storage batteries	M3.03	R
9	Draw simple schematics of an Electric Vehicle, and explain the operations	M4.01	R
OR			
10	Differentiate Slow charger and fast charger	M4.02	R
11	Outline the characteristics of the slow charging method	M4.02	U
OR			
12	Explain communication methodology of DC fast chargers	M4.03	U